FOUR EQUIVALENT PREPARATIONS FOR THE EXISTENCE OF NORMALIZATIONS OF AN ALGEBRAIC STACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The corresponding conditions for morphisms of algebraic spaces. Invoked inside the counted proof (source lines 10218-10224) as cross-chapter pointers. Supplies the space-level formulation of the condition.
- The statement that the number of irreducible components is local in the smooth topology. Invoked at the start of the counted proof (source lines 10218-10221). Makes the conditions (1) to (3) equivalent.
- The presentation of a quasi-compact stack by an affine scheme. Invoked inside the counted proof (source lines 10228-10230) as a cross-chapter pointer. Produces the affine scheme whose finitely many components bound those of the test object.

Source reference. Original reference descent-lemma-locally-finite-nr-irred-local-fppf: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10219–10219. Navigation pointer only: the counted text refers to the Stacks item descent-lemma-locally-finite-nr-irred-local-fppf, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-prepare-normalization: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10209–10209. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-prepare-normalization, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference stacks-properties-lemma-quasi-compact-stack: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1935–1935. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-quasi-compact-stack, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-type-property: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10220–10220. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-type-property, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference topology-lemma-surjective-continuous-irreducible-components: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10232–10232. Navigation pointer only: the counted text refers to the Stacks item topology-lemma-surjective-continuous-irreducible-components, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. The following are equivalent*

1. *there is a surjective smooth morphism $U \to \mathcal{X}$ where U is a scheme such that every quasi-compact open of U has finitely many irreducible components,*
2. *for every scheme U and every smooth morphism $U \to \mathcal{X}$ every quasi-compact open of U has finitely many irreducible components,*
3. *for every algebraic space Y and smooth morphism $Y \to \mathcal{X}$ the space Y satisfies the equivalent conditions of Morphisms of Spaces, Lemma spaces-morphisms-lemma-prepare-normalization, and*
4. *for every quasi-compact algebraic stack $\mathcal{Y}$ smooth over $\mathcal{X}$ the space $|\mathcal{Y}|$ has finitely many irreducible components.*

Proof. The equivalence of (1), (2), and (3) follow from Descent, Lemma descent-lemma-locally-finite-nr-irred-local-fppf, Properties of Stacks, Lemma stacks-properties-lemma-type-property, and Morphisms of Spaces, Lemma spaces-morphisms-lemma-prepare-normalization. It is also clear from these references that condition (4) implies condition (1). Conversely, assume the equivalent conditions (1), (2), and (3) hold and let $\mathcal{Y} \to \mathcal{X}$ be a smooth morphism of algebraic stacks with $\mathcal{Y}$ quasi-compact. Then we can choose an affine scheme V and a surjective smooth morphism $V \to \mathcal{Y}$ by Properties of Stacks, Lemma stacks-properties-lemma-quasi-compact-stack. Since V has finitely many irreducible components by (2) and since $|V| \to |\mathcal{Y}|$ is surjective and continuous, we conclude that $|\mathcal{Y}|$ has finitely many irreducible components by Topology, Lemma topology-lemma-surjective-continuous-irreducible-components. □